prefer to focus on each stage of the preprocessing, training and deployment processes. For instance, Amazon and Google's cloud platforms offer a set of endpoints to address machine learning on streaming data. In fact, their recent offerings like Google Cloud AutoML and the Amazon Web Services SageMaker focus on transferring control into the user's hands by introducing more interpretability; but they still have some distance to travel when considering the level of automation and performance across heterogenous data sets.

Following Rekognition in 2017, Amazon has ramped up focus on natural language processing, automatic speech recognition, text-to-speech services, and neural machine translation technologies as managed services in the cloud. The company introduced video and image analysis using DeepLens, making it easier for developers to access these as desired.

Figure 1: Artificial Intelligence: A force of good and evil (Credits: *Hackernoon*)

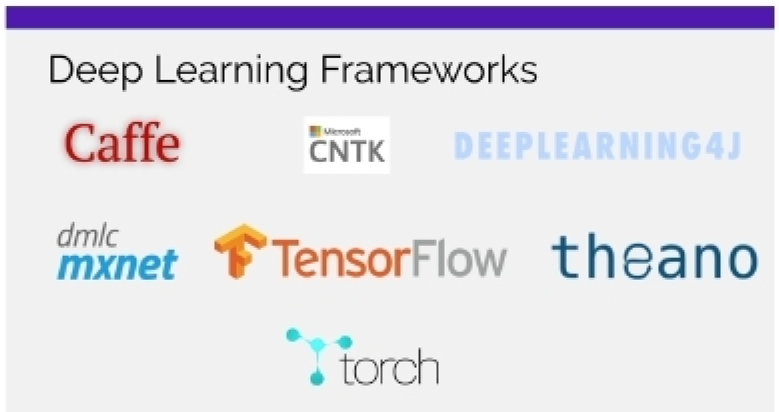

Figure 2: AI/ML tools (Credits: *Amazon*)


Figure 3: Machine learning on mobile devices (Credits: *Veltrod*)

Top Web frameworks for machine learning

Web frameworks have been all the rage in machine learning as neural networks have reduced in size and gained sufficient accuracy when compared to human standards. One of the forerunners in this space is Andrej Karpathy's ConvNet.js. This inspired similar work or parallel lines of thought that later resulted in libraries based in JavaScript, which can be run as part of server-side or client-side scripts. The recent release of TensorFlow.js is a solid step forward in this direction, extending the ecosystem for developers seeking to bring the machine learning experience to the browser. There are other frameworks, including ml5js, focused on offering a complete set of in-browser machine learning capabilities.

Top ML tools for mobile app developers

'Data is more valuable than gold these days'. All mobile app developers want to integrate advanced analytics including machine learning systems into data processing, to enable them to generate more accurate insights and make decisions based on the 'big picture' of the user statistics within their apps. It becomes a very lucrative market to capture as app developers seek custom solutions for their use cases, which focus on an unchanged user experience in spite of a huge amount of processing in the backend.

Google is capitalising on its expertise in developing and supporting TensorFlow by releasing ML Kit which caters to the Android market, often with specific requirements of low-memory impact and low-resource learning, on the fly. This comprises specific libraries that address text and face recognition, bar code scanning, image labelling and face detection, and will soon see a foray into natural language processing, with support for the smart reply feature seen in its other products including Gmail.

Apple, meanwhile, is playing catch-up with its CoreML library. The advantage of the competition within this space has been the release of a huge repository of lightweight models optimised for mobile devices that permit the end user to continue to have a streamlined experience on handheld devices.

Overall, the machine learning landscape is growing increasingly crowded with new frameworks emerging and older ones fading; but a huge amount of work is focused on working in conjunction and promoting a much healthier environment for developers and researchers. This bodes well for the end users as we note the emergence of mature, capable and interpretable open source machine learning tools for use cases spanning the tech landscape and beyond. END

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